

THEORETICAL TASKS

1. Explain the End-to-End Data Analysis Lifecycle using a Real Business Example

Business Example: An e-commerce company wants to reduce product return rates.

1. Business Understanding:

The company identifies a business problem, i.e., high product return rate.

2. Problem Definition:

The objective is defined clearly: reduce return rate by 15% within three months.

3. Data Collection:

Data is collected from different sources such as sales records, return logs, customer feedback, and product catalogs.

4. Data Cleaning:

Errors, duplicate values, and missing data are removed or corrected.

5. Data Integration:

Multiple datasets (sales, returns, reviews) are merged into a single dataset.

6. Exploratory Data Analysis (EDA):

Patterns and trends are identified, such as which products have the highest return rates.

7. Feature Selection:

Important variables such as product category, size, price, and delivery time are selected for analysis.

8. Modeling / Statistical Analysis:

Predictive or statistical models are applied to identify factors influencing returns.

9. Visualization and Reporting:

Dashboards and reports are created to present findings to stakeholders.

10. Insight Generation:

Insights such as “incorrect size charts lead to higher returns” are derived.

11. Decision Making:

Business actions are taken, such as improving size guides and product descriptions.

12. Monitoring and Evaluation:

The impact of decisions is tracked over time.

13. Feedback Loop:

New data is continuously used to improve future analysis.

2. Differentiate

(a) Descriptive Analytics vs Diagnostic Analytics

Descriptive Analytics	Diagnostic Analytics
1. Focuses on what happened.	1. Focuses on why it happened.
2. Summarizes historical data.	2. Analyzes reasons behind trends.
3. Uses reports and dashboards.	3. Uses drill-down and root cause analysis.
4. Example: Total sales last month.	4. Example: Reason for sales drop.
5. Uses averages and totals.	5. Uses correlations and comparisons.

- 6. Simple and basic analysis. | More detailed and complex analysis.
- 7. Answers “What?”. | Answers “Why?”.
- 8. First stage of analytics. | Second stage of analytics.
- 9. Helps understand past performance. | Helps understand problems.
- 10. Supports monitoring activities. | Supports problem-solving activities.

(b) Correlation vs Causation

| Correlation | Causation |

| ----- | ----- |

- 1. Shows relationship between variables. | Shows cause-and-effect relationship.
- 2. Variables move together. | One variable directly affects another.
- 3. Does not imply reason. | Explains the reason.
- 4. Example: Ice cream sales and temperature. | Temperature causes ice cream sales.
- 5. Can be accidental. | Must be logically proven.
- 6. Used in exploratory analysis. | Used in decision making.
- 7. Can be positive or negative. | Always directional.
- 8. Needs further testing. | Needs experiments or strong evidence.
- 9. Risk of false interpretation. | More reliable conclusion.
- 10. Means “associated”. | Means “responsible”.

3. Short Notes

(a) Data Bias

- 1. Data bias occurs when data is not representative of the population.
- 2. It leads to incorrect and unfair results.

3. It can occur due to poor sampling methods.
4. Example: Collecting data only from urban customers.
5. It affects machine learning model accuracy.
6. It creates ethical and social issues.
7. It misguides business decisions.
8. It reduces reliability of analysis.
9. It is difficult to detect without domain knowledge.
10. It can be minimized using diverse and balanced data sources.

(b) Missing Data Strategies

1. Remove rows or columns with missing values.
2. Replace missing values using mean, median, or mode.
3. Use forward fill or backward fill methods.
4. Apply interpolation techniques.
5. Predict missing values using machine learning models.
6. Treat missing values as a separate category.
7. Analyze the pattern of missingness.
8. Avoid deleting important observations blindly.
9. Use domain knowledge to fill values.
10. Choose the strategy based on data type and business context.

(c) KPIs vs Metrics

1. KPIs (Key Performance Indicators) measure critical business goals.
2. Metrics measure general business activities.
3. KPIs are strategic in nature.
4. Metrics are operational in nature.
5. KPIs are directly linked to objectives.

6. Metrics support KPIs.
7. Example KPI: Customer retention rate.
8. Example Metric: Number of daily active users.
9. KPIs are few and focused.
10. Metrics can be many and detailed.

4. Case: “Why do dashboards fail even with correct data?”

1. Dashboards are not aligned with business objectives.
2. Too many charts create confusion.
3. Incorrect visualization types are used.
4. Key KPIs are not clearly defined.
5. Data is not refreshed regularly.
6. Users are not trained to interpret dashboards.
7. Poor layout and color selection reduce readability.
8. Dashboards lack insights and storytelling.
9. Too much technical information is shown.
10. No action-oriented recommendations are provided.
11. Stakeholders do not trust or adopt the dashboard.
12. Dashboards are not role-specific.
13. Interactivity such as filters and drill-down is missing.
14. Raw data is shown instead of summarized insights.